CLASS 9 — SCIENCE · FORMULA SHEET

Chapter 9: **Cell: The Building Block of Life** | Quick revision, one page

Cell theory

- Schleiden and Schwann: all plants and animals are made of cells; the cell is the basic unit of life. Virchow: new cells arise from pre-existing cells.
- Robert Hooke observed and named cells in cork (1665). Unicellular organisms: *Amoeba*, *Paramecium*, bacteria. Multicellular organisms: plants, animals.

Prokaryotic and eukaryotic cells

- Prokaryotic: no true nucleus (no nuclear membrane), DNA free in the cytoplasm, no membrane-bound organelles (bacteria).
- Eukaryotic: true nucleus with nuclear membrane, membrane-bound organelles (plants, animals, fungi, protists).
- Sizes: $1\ \mu\text{m} = 10^{-6}\ \text{m}$; bacteria about 1 to 10 μm; the human egg cell is about 100 μm.

Cell structures and functions

- Cell membrane: selectively permeable, controls entry and exit. Cell wall (plants, fungi, bacteria): rigid, protective; cellulose in plants.
- Nucleus: control centre, holds DNA (chromosomes). Cytoplasm: jelly-like medium. Ribosomes: protein synthesis.
- Rough ER: protein synthesis and transport; smooth ER: lipid synthesis and detoxification. Golgi apparatus: modifies, packs and dispatches materials.
- Mitochondria: respiration, ATP production. Chloroplasts (plastids): photosynthesis, contain chlorophyll. Lysosomes: digestion, 'suicide bags'. Vacuoles: storage, turgidity.

Transport across the membrane

- Diffusion: movement of molecules from higher to lower concentration. Osmosis: diffusion of water through a selectively permeable membrane.
- Hypotonic medium: cell gains water and swells. Hypertonic medium: cell loses water and shrinks (plasmolysis). Isotonic: no net movement of water.

Plant cell vs animal cell

- Plant cell: cell wall, chloroplasts, large central vacuole. Animal cell: no cell wall or chloroplast, small vacuoles, centrioles.
- Mass change in osmosis (per cent) $= \dfrac{\text{final mass} - \text{initial mass}}{\text{initial mass}} \times 100$.